

East Point College of Higher Education

Placement Training Class Report

Name : Mithun Chakravarthi S G

Reg No : P19MT25S126004

Class: 1st Sem MCA

Multiplication of Table Code

```
C: > Users > gokscst > htmlprograms > m.py > ...  
1  n=int(input("enter number:"))  
2  i=1  
3  while i<=10:  
4      print(n*i)  
5      i+=1
```

Output

```
PS C:\Users\gokscst> & "C:/Program Files/Python312/python.exe" c:  
enter number:1  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
PS C:\Users\gokscst> |
```

Code

```
for i in range(2,10,2):  
    print(i)
```

Output

```
/Users/gokscst/htmlprograms/m.py  
2  
4  
6  
8  
PS C:\Users\gokscst>
```

Code to get Average

```
def average(l):  
    sum = 0  
    for x in l:  
        sum=sum+x  
    n=len(l)  
    return sum/n  
l=[10,20,30,40]  
print(average(l))
```

Output

```
PS C:\Users\gokscst> & "C:/Program Files/Python312/python.exe" c:/Users/gokscst/htmlprograms/ml.py 25.0
PS C:\Users\gokscst> 
```

Code to Get Samellet element in List

```
def getsmaller(l,x):
    res=[]
    for e in l:
        if e<x:
            res.append(e)
    return res
l=[8,100,20,40,3,7]
x=10
print(getsmaller(l,x))
```

Output

```
NameError: name 'x' is not defined. Did you mean: 'X'?
PS C:\Users\gokscst> & "C:/Program Files/Python312/python.exe" c:/Users/gokscst/htmlprograms/ml.py
[8, 3, 7]
PS C:\Users\gokscst> 
```

Even or Odd Number from List

```
numbers=[10,41,30,15,80]
evens=[n for n in numbers if n % 2==0]
odds=[n for n in numbers if n % 2!=0]
print(f"Even={evens}")
print(f"Odd={odds}")
```

Output

```
Even=[10, 30, 80]
Odd=[41, 15]
PS C:\Users\gokscst> 
```

Remove Duplicates Code

```
class ALGO04twoPointerRemoveDuplicates:
    def remove_duplicates(self,nums):
        if not nums:
            return 0
        left = 0
        for right in range(1,len(nums)):
            if nums[left]!=nums[right]:
                left += 1
                nums[left]=nums[right]
        return left +1
if __name__ == "__main__":
    solution = ALGO04twoPointerRemoveDuplicates()
    nums=[1,2,3,3,3,4,4,5]
    unique_index = solution.remove_duplicates(nums)
    print(unique_index)
    print(nums[:unique_index])
```

Output

```
[1, 2, 3, 4, 5]
PS C:\Users\gokscst>
```

Code for to Check Palindrome or not

```
s=input("Enter a string:")
low=0
high=len(s)-1
while(low<high):
    if s[low] != s[high]:
        print("No")
        break
    low=low+1
    high=high-1
else:
    print("Yes")
```

Output:

```
PS C:\Users\ADMIN> D:/ProgramData
PS C:\Users\ADMIN> conda activate
PS C:\Users\ADMIN> & D:/ProgramData
Enter a string:tamil
No
PS C:\Users\ADMIN> □
```

Code for TwoSum

```
class Solution:
    def twoSum(self, nums: List[int], target: int) -> List[int]:
        prevmap={}
        for i, n in enumerate(nums):
            diff=target - n
            if diff in prevmap:
                return [prevmap[diff], i]
            prevmap[n]=i
```

Output:

Accepted Runtime: 0 ms

✓ Case 1

✓ Case 2

✓ Case 3

Input

nums =
[2,7,11,15]

target =
9

Output

[0,1]

Expected

[0,1]

Accepted Runtime: 0 ms

✓ Case 1

✓ Case 2

✓ Case 3

Input

```
nums =  
[3,2,4]
```

```
target =  
6
```

Output

```
[1,2]
```

Expected

```
[1,2]
```

Code for Anagram

```
word1 = "listen"  
word2 = "silent"  
  
if (sorted(word1) == sorted(word2)):  
    print(f'"{word1}" and "{word2}" are anagrams.' )
```

Output:

```
PS C:\Users\gokscst> & "C:/Program F:  
"listen" and "silent" are anagrams.  
PS C:\Users\gokscst> [
```

Binary Search Code

```
def binary_search(arr,target):
    left,right = 0,len(arr)-1
    while left <= right:
        mid = left +(right - left) //2
        if target == arr[mid]:
            return mid
        if arr[mid] < target:
            left = mid +1
        elif arr[mid] > target:
            right = mid -1
    return -1
data =[2,4,9,99,1000]
target = 1000
target_index= binary_search(data,target)
print(target_index)
```

Output:

```
PS C:\Users\gokscst> & "C:/Program Files/Python312/python.exe" -i
4
PS C:\Users\gokscst>
```

Code for to Find Prefix of a String

```
def longest_common_prefix(strs):  
    if not strs:  
        return ""  
  
    prefix = strs[0]  
  
    for s in strs[1:]:  
        while not s.startswith(prefix):  
            prefix = prefix[:-1]  
            if not prefix:  
                return ""  
    return prefix  
  
print(longest_common_prefix(["abc", "ab", "a"]))
```

Output:

```
PS C:\Users\gokscst> & "C:/Program Files/Python312/python.exe"  
a  
PS C:\Users\gokscst> 
```

